MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, SYNTHETIC FIBER REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (LOW PRESSURE SIZES -40, -48 and -64) AND ADAPTERS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic fiber reinforced hoses and fittings in sizes -40 through -64 for low pressure applications.

1.2 Classification. End fittings for hose covered by this specification shall be in sizes -40 through -64 as applicable, and shall be of the following types:

Type AI - In accordance with figure 1, with 400 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AII - In accordance with figure 1, with 150/250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AIII - In accordance with figure 1, with 250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AIV - In acccordance with figure 1, with 150 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type AV - In accordance with figure 1, with 150 lb/in^2 B16.5 flange (see figure 1 of MIL-H-24136).
Type BI - In accordance with figure 2, with 400 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BII - In accordance with figure 2, with 150/250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BIII - In accordance with figure 2, with 250 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BIV - In accordance with figure 2, with 150 lb/in^2 flange (see figure 1 of MIL-H-24136).
Type BV - In accordance with figure 2, with 150 lb/in^2 B16.5 flange (see figure 1 of MIL-H-24136).
Type C - In accordance with figures 3 and 6 or 7 and 8.
Type D - In accordance with figures 4, 6 or 7 and 8.
Type E - In accordance with figure 5.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Unless otherwise specified, the following specifications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-654 - Brazing Alloys, Silver.
QQ-C-390 - Copper Alloy Castings, (Including Cast Bar).
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-H-24136 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification For.
MIL-B-24480 - Bronze, Nickel-Aluminum Castings, for Seawater Service.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Hose and end fittings furnished under this specification shall conform to the requirements of MIL-H-24136 and as specified herein.

3.2 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

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TABLE I. Dash number, performance characteristics and dimensions.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)				Hose reinforcement		Minimum bend radius (inches)	Peak impulse pressure (lb/in^2)
	Working (max)	Burst (min)	I.D.		O.D.		Construction	No. of layers		
			Min	Max	Min	Max				
-40	1000	4000	2.454	2.531	3.230	3.360	Spiral	6	20	1500
-48	900	3600	2.954	3.031	3.730	3.860	Spiral	6	24	1350
-64	600	2400	3.938	4.062	4.800	5.050	Spiral	6	30	900

3.3 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24136 and as specified herein.

4.2 Sampling for qualification testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings from at least two previously qualified suppliers (see 4.2.4).

TABLE II. Number of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-40	3
-64	4

4.2.2 End fittings. Qualification testing of end fittings only, shall consist of 14 end fittings of the same type as specified in table III. End fitting samples shall be installed on seven qualified hose samples, in the sizes listed in table II, from at least two previously qualified hoses (see 4.2.4).

TABLE III. Number of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-40	6
-64	8

4.2.3 Hose and end fittings combined. In the event that a single contractor of both hose and fittings desires to qualify both hose and fittings, or a hose contractor and a fitting contractor desire to qualify hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End Fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-40	2	1	-40	4	2
-64	2	2	-64	4	4

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2		Qualifier 2/	Approved 1/ source #1	Approved 1/ source #1
-40	2	1	0	-40	4	2	0
-64	2	1	1	-64	4	2	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

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4.2.4 Sample requirements when there are less than two previously qualified suppliers. Where there is only one qualified source for qualified samples all the previously qualified samples required shall come from that source. Where there are no previously qualified sources, all samples of hose and fittings shall be submitted by the prospective qualifier.

4.2.5 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

Test sequence	Sample numbers						
	Size (dash no.) -40			Size (dash no.) -64			
	1	2	3	4	5	6	7
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X		X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X			X		
Impulse test (see 4.6.5)1/			X				X
Burst test (see 4.6.6)1/	X			X			
Vacuum test (see 4.6.7)1/			X				X
Cold temperature flexing (see 4.6.8)1/			X			X	
Adhesion test (see 4.6.9)1/			X				X

1/ See MIL-H-24136.

4.2.6 Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III and IV will qualify the manufacturer for all sizes shown in table I.

5. PACKAGING

5.1 Preparation for delivery requirements shall be in accordance with MIL-H-24136.

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6. NOTES

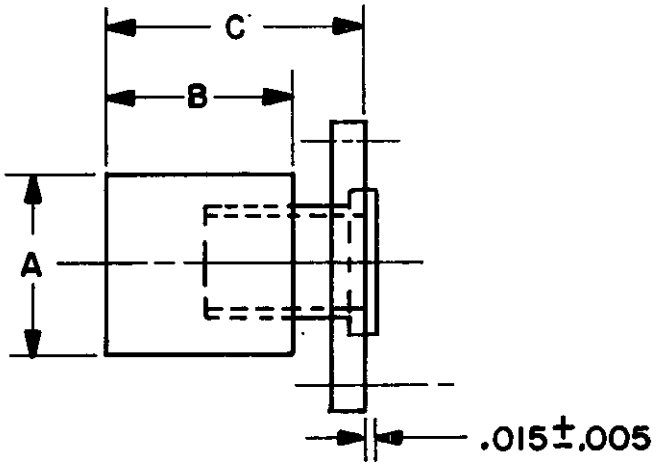
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard or shore to ship piping systems where pressures do not exceed the rated working pressure and temperatures do not exceed minus 40°F or 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4730-N874-4)

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Hose size (dash no.)	A 1/	B 1/	C 1/
-40	6.75	5.00	5.75
-48	7.75	5.00	5.875
-64	9.00	5.00	6.00

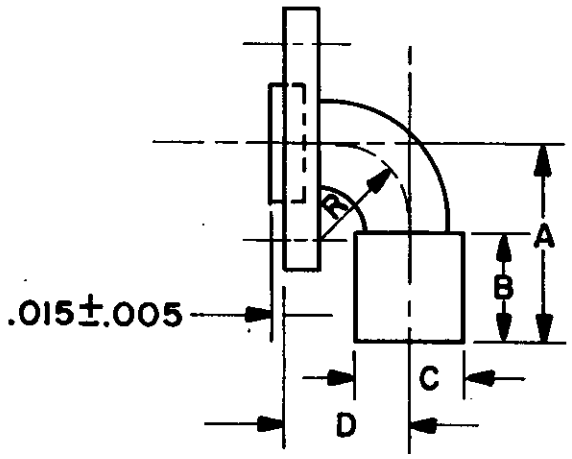
1/ Maximum dimensions.

SH 12307

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-281, QQ-N-286.
   All other parts - QQ-N-281, QQ-N-286, QQ-N-288 or MIL-B-24480.

FIGURE 1. Type AI, AII, AIII, AIV or AV hose to flange, straight.

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Hose size (dash no.)	A (nom)	B 1/	C 1/	D (nom)	R (nom)
-40	6.25	5.00	6.75	4.50	2.125
-48	6.875	5.00	7.75	5.00	2.50
-64	7.00	5.00	9.00	5.75	3.00

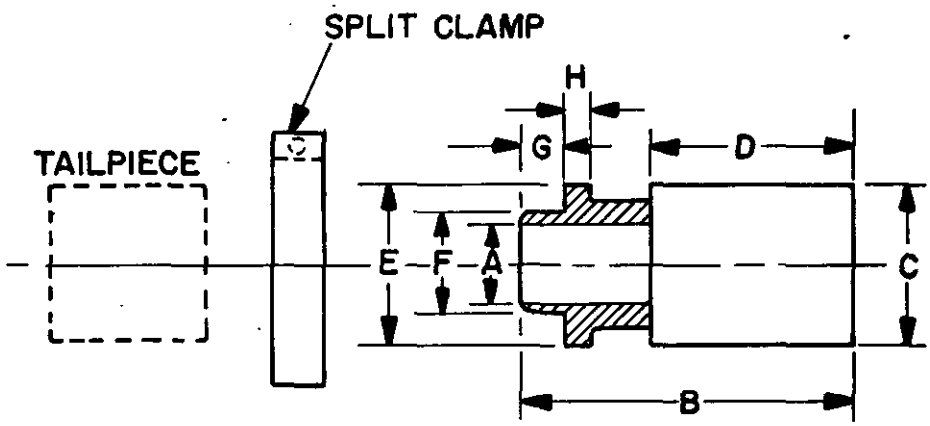
1/ Maximum dimensions.

SH 12308

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
Bolts and nuts - QQ-N-281, QQ-N-286.
All other parts - QQ-N-281, QQ-N-286, QQ-N-288, or MIL-B-24480.

FIGURE 2. Types BI, BII, BIII, BIV or BV hose to flange, 90 degree elbow.

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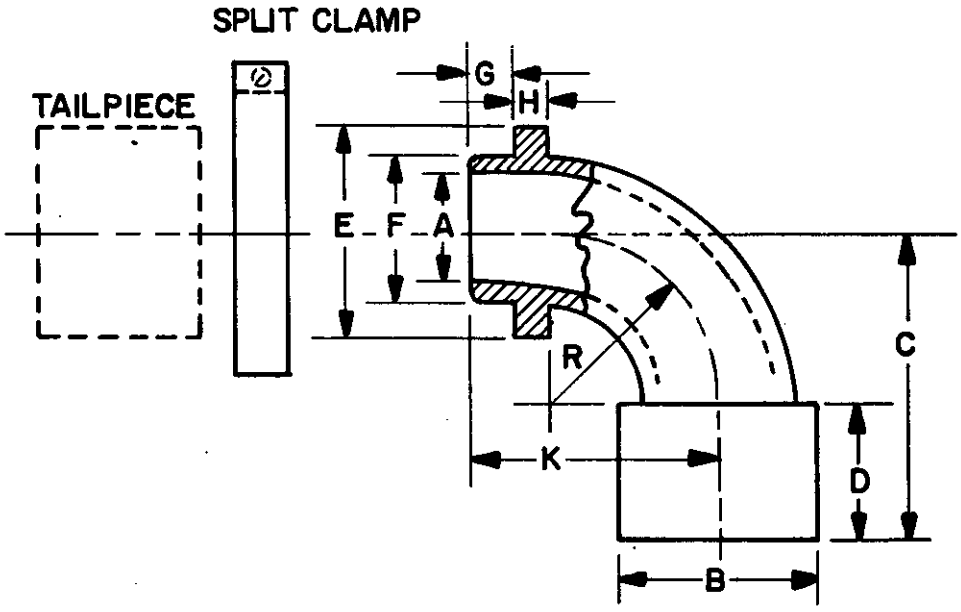
Hose size (dash no.)	A (nom)	B (max)	C (max)	D (max)	E +0.006	F +0.006 -0.000	G +0.005 -0.000	H +0.005 -0.000
-40	2	6.188	6.75	5.00	3.245	2.495	0.332	0.317
-48	2-1/2	6.188	7.75	5.00	3.745	2.995	0.332	0.317
-48	3	6.188	7.75	5.00	4.370	3.620	0.332	0.317
-64	3-1/2	6.313	9.00	5.00	4.928	4.120	0.332	0.317
-64	4	6.313	9.00	5.00	5.428	4.620	0.404	0.360

SH 12309

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-281, QQ-N-286.
   All other parts - QQ-N-281, QQ-N-286 or QQ-N-288.

FIGURE 3. Type C, straight split clamp with braze tailpiece and clamp assembly.

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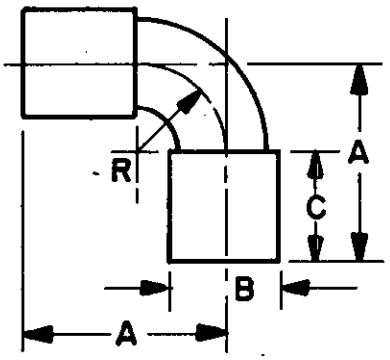
Hose size (dash no.)	A (max)	B (min)	C (max)	D (max)	E ±0.006	F +0.006 -0.000	G +0.005 -0.000	H +0.005 -0.000	R (nom)	K (nom)
-40	2	6.75	6.25	5.00	3.245	2.495	0.332	0.317	2.125	3.50
-48	2-1/2	7.75	6.875	5.00	3.745	2.995	0.332	0.317	2.50	3.75
-48	3	7.75	6.875	5.00	4.370	3.620	0.332	0.317	2.50	3.75
-64	3-1/2	9.00	7.00	5.00	4.928	4.120	0.332	0.317	3.00	4.50
-64	4	9.00	7.00	5.00	5.428	4.620	0.404	0.360	3.00	4.50

SH 12310

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 4. Type D, split clamp 90 degrees.

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Hose size (dash no.)	A (nom)	B 1/	C 1/	R (nom)
-40	6.25	6.75	5.00	2.125
-48	6.875	7.75	5.00	2.50
-64	7.00	9.00	5.00	3.00

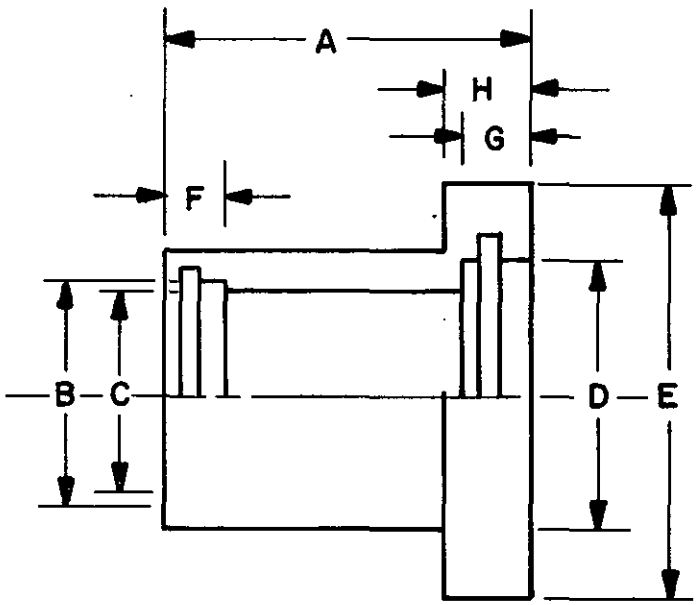
1/ Maximum dimensions.

SH 12311

NOTES:
1. Dimensions are in inches.
2. Materials: All parts in direct contact with sea water shall be in accordance with MIL-B-24480.
   Bolts and nuts - QQ-N-281, QQ-N-286.
   All other parts - QQ-N-281, QQ-N-288, MIL-B-24480 or QQ-N-286.

FIGURE 5. Type E, 90 degree elbow.

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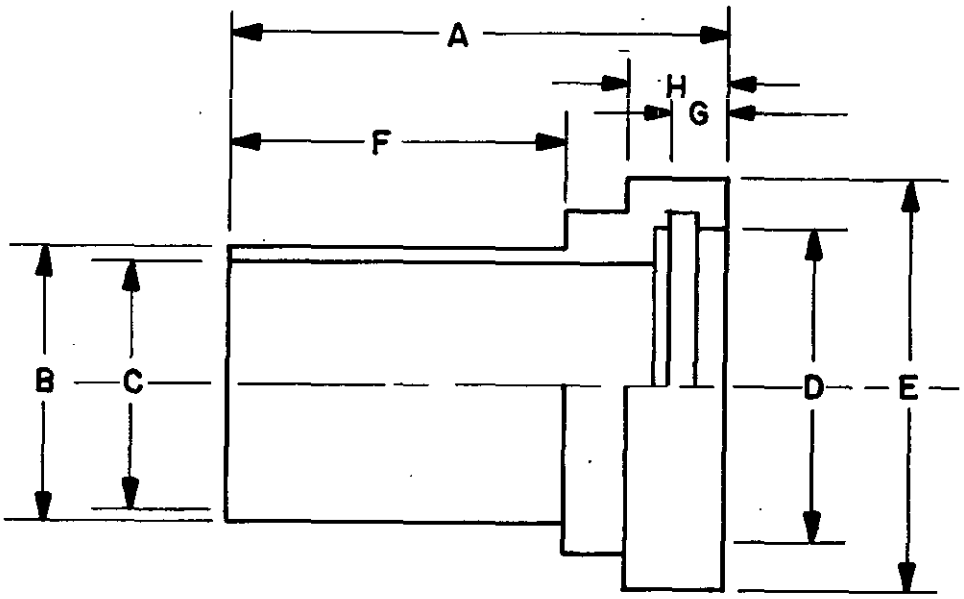
Hose size (dash no.)	A (max)	B +0.006 -0.000	C (nom)	D +0.006 -0.000	E ±0.006	F (nom)	G +0.005 -0.000	H +0.005 -0.000
-40	1.86	2.88	2.59	3.01	3.75	0.78	0.35	0.317
-48	1.55	3.50	3.33	3.64	4.36	0.83	0.35	0.317
-64	1.70	4.50	4.31	4.64	5.43	0.91	0.42	0.36

SH 12312

NOTES:
1. Dimensions are in inches.
2. Material: MIL-B-24480
   O-ring - buna N
   Braze ring - QQ-B-654

FIGURE 6. Sil-braze tailpiece, female.

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Hose size (dash no.)	A (max)	B +0.006 -0.000	C (nom)	D +0.006 -0.000	E +0.006 -0.000	F (nom)	G +0.005 -0.000	H +0.005 -0.000
-40	2.04	2.88	2.59	3.01	3.75	1.12	0.35	0.37
-48	3.45	3.50	3.00	3.63	4.36	2.50	0.35	0.37
-64	3.67	4.50	3.88	4.63	5.43	2.64	0.42	0.36

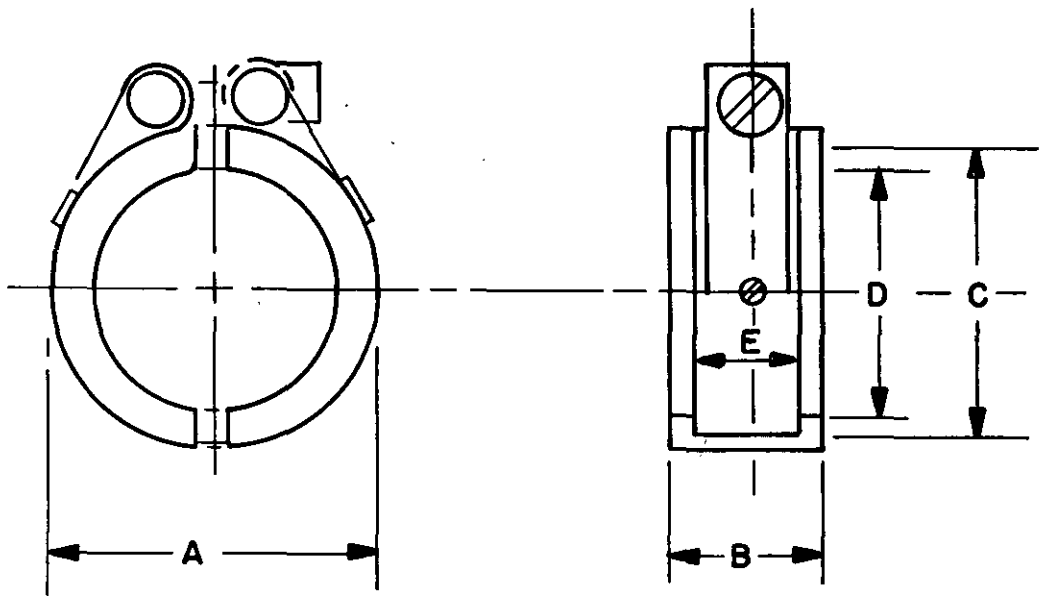
SH 12313

NOTES:
1. Dimensions are in inches.
2. Material: MIL-B-24480
   O-ring - buna N

FIGURE 7. Male sil-braze tailpiece.

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Hose size (dash no.)	A (max)	B (max)	C +0.006 -0.000	D (nom)	E +0.006 -0.000
-40	4.42	1.44	3.75	3.25	0.674
-48	5.19	1.50	4.36	3.86	0.624
-64	6.12	1.62	5.43	4.93	0.720

SH 12314

NOTES:
1. Dimensions are in inches.
2. Material: Split clamp - QQ-C-390, alloy 955
   Band - QQ-N-281 and QQ-C-591, alloy 655
   Screws - QQ-C-591, alloy 655

FIGURE 8. Split clamp assembly.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24136/4(SH)	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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